Presented by :
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NTHU iGEM

AeroSense: Turning smell into data

THE PROBLEM

Spoilage smells before it shows

Diseased produce releases volatile organic compounds (VOCs) days to weeks before any visible symptom. Today's tools miss that window.

Visual inspection
Fast, but sees only symptoms. By then the batch has often shipped.

Lab assays (PCR, GC-MS)
Accurate, but slow, destructive and off-site.

TAIWAN BANANA SECTOR

NT$3.5B

estimated yearly fungal-disease loss risk, about a third of the sector's NT$10.4B output

OUR SOLUTION

Borrow the fruit fly's nose and brain

01 · WET LAB

Nose

Fly olfactory receptors in human cells turn odor into light

02 · MODEL

Brain

A fly mushroom-body circuit turns light into a decision

03 · HARDWARE

Device

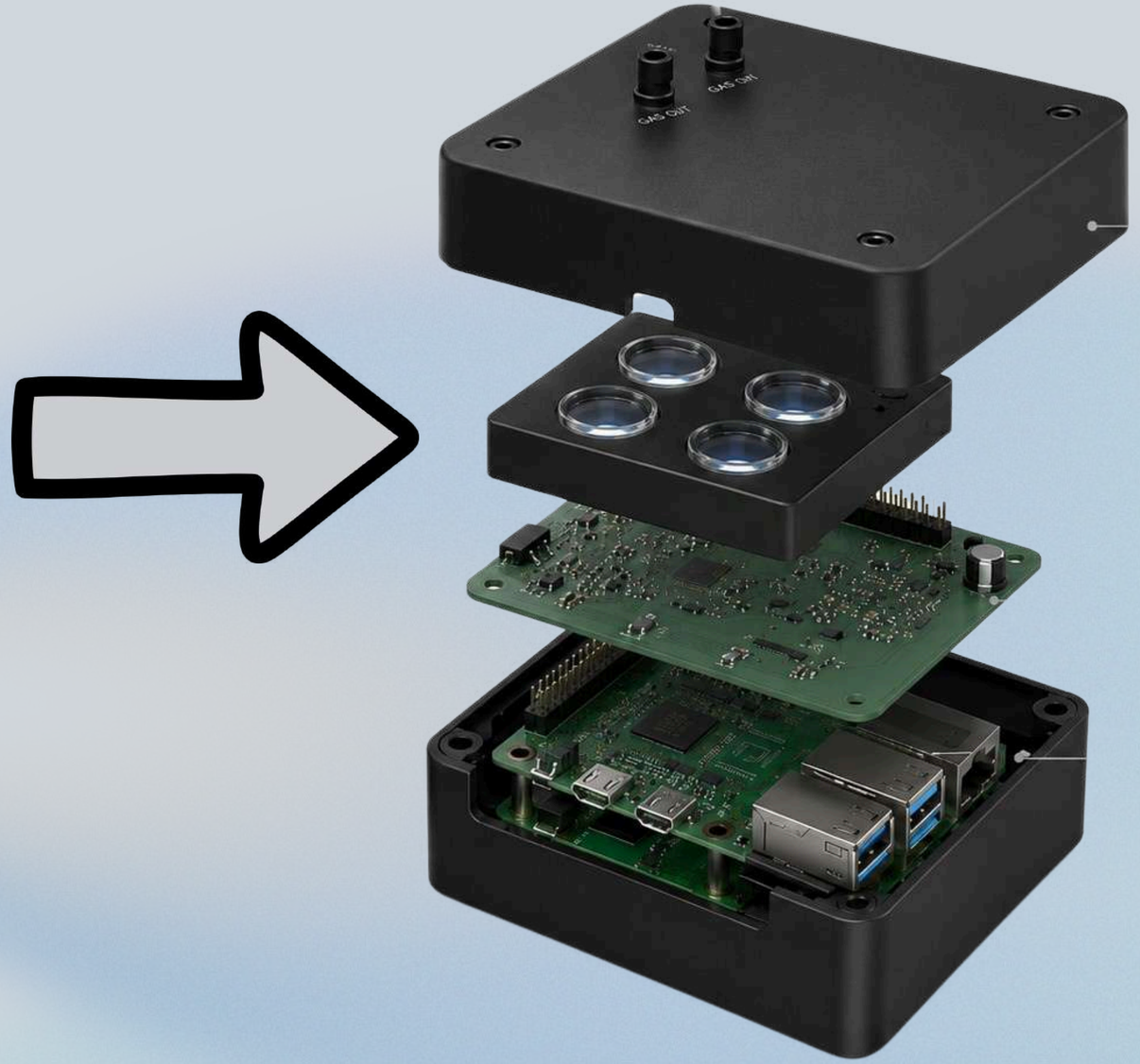
Lock-in optics pull a faint glow out of real-world noise

Human Practices shaped every layer

WET LAB

Can a living cell smell?




GAS OUT
GAS OUT

WET LAB · SENSOR DESIGN

A living cell that glows when it smells

1
VOC binds
to a fly olfactory receptor (OR)

→

2
Channel opens
OR + Orco form an ion channel

→

3
Ca^{2+} flows in
calcium enters the cell

→

4
Cell glows
GCaMP6 fluoresces: stronger smell, brighter cell

Chassis: HEK293T
A well-characterized human cell line. No live insects at any point.

Self-renewing
Cells keep making fresh receptors, so there is no chemical film to age or drift.

WET LAB · RECEPTOR LIBRARY

Seven receptors, two early leads

Candidate ORs chosen from a ligand-profiling study of plant and plant-pathogen volatiles

Or10a · **Or13a** · Or47a · **Or49b** · Or71a · Or85b · Or98a

Zboray et al., 2023, Scientific Reports

BACTERIAL LEAD

Or49b → indole

Indole is a classic marker of E. coli contamination. Or49b is already cloned and sequenced.

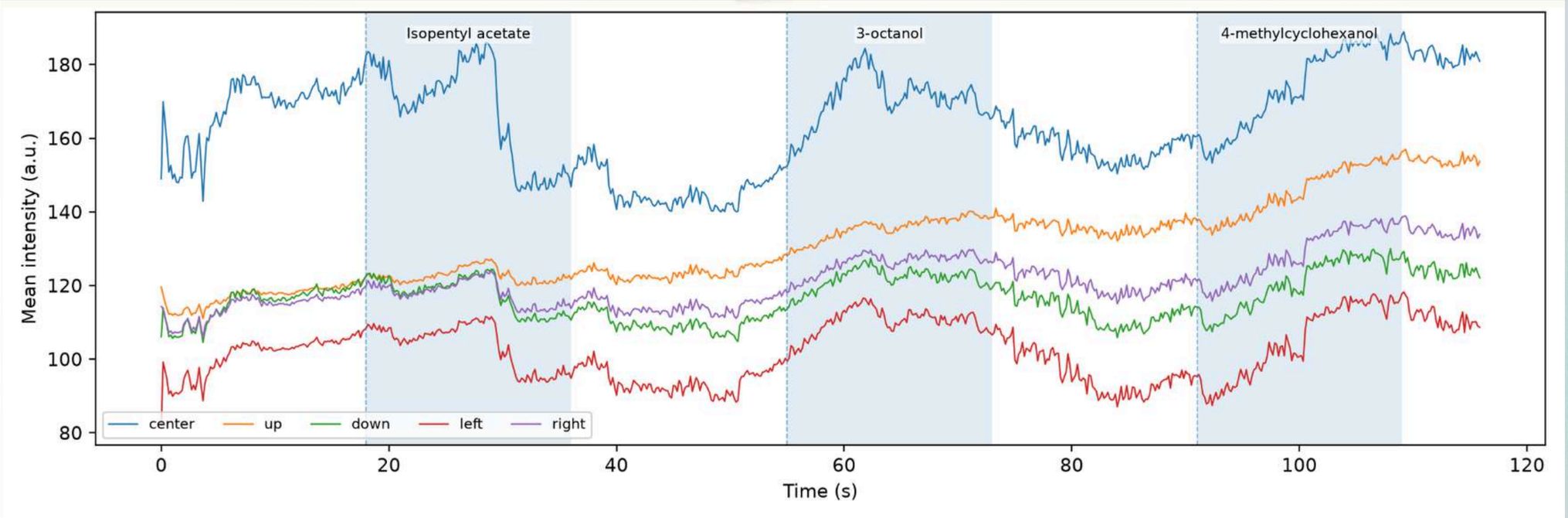
FUNGAL LEAD

Or13a → 1-octen-3-ol

"Mushroom alcohol", a fungal volatile, fits our crop-disease focus.

WET LAB · RECEPTOR LIBRARY

Seven receptors, two early leads



Readout: GCaMP fluorescence, 5 regions

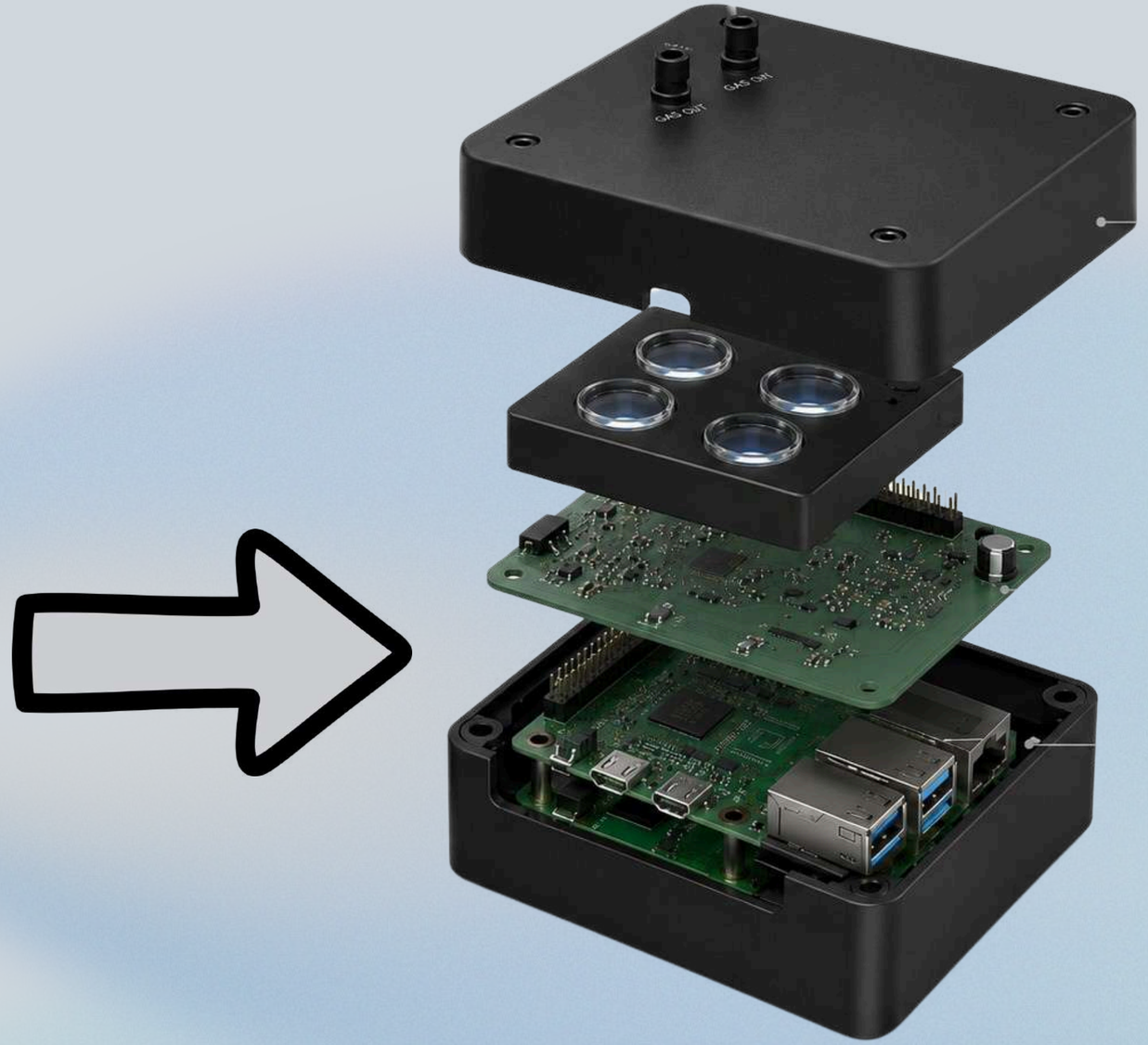
Clearest rise: 3-octanol window

Next: $\Delta F/F_0$, solvent control, target VOCs

MODEL

The cell lit up. What did it smell?




GAS OUT
GAS OUT

MODEL · NEUROMORPHIC DECODER

Decoding with a fly's brain

PN

PROJECTION NEURONS

Lateral inhibition boosts signal-to-noise

→

KC

KENYON CELLS

Sparse coding separates overlapping odors

→

MBON

OUTPUT NEURONS

Make the call: which odor, and is it a risk?

Built on our advisors' own published work

Cheng, Charng, … Lo, Lee (2025). Hybrid neural networks in the mushroom body drive olfactory preference in Drosophila. Science Advances 11(22).

MODEL · WHY A FLY BRAIN

Small enough to run at the edge

<16 ms

decoding latency per reading

<1 W

sensing to classification on a Raspberry Pi

>85%

single-VOC accuracy: **design target**, not yet measured

HARDWARE

Can we hear a faint glow in a noisy world?



HARDWARE · ENGINEERING ITERATION

Why we replaced the camera

V1 · CAMERA

Blue LED + CMOS, 30 fps

Works on the bench. Outside the lab, ambient light and sensor noise swamp a faint cell signal.

V2 · LOCK-IN

Photodiode + DLIA

Modulated LEDs with digital lock-in amplification: read only the frequency we sent, ignore the rest.

Industry told us the same thing: **sensitivity that holds in the lab often collapses in the field.**

HARDWARE · DIGITAL LOCK-IN AMPLIFICATION

Listening for one frequency

4 LEDs
each blinks at its own frequency, f1–f4

→

Cells
fluoresce in step with each LED

→

Photodiode
one detector collects all light

→

TIA + ADC
amplify and digitize

→

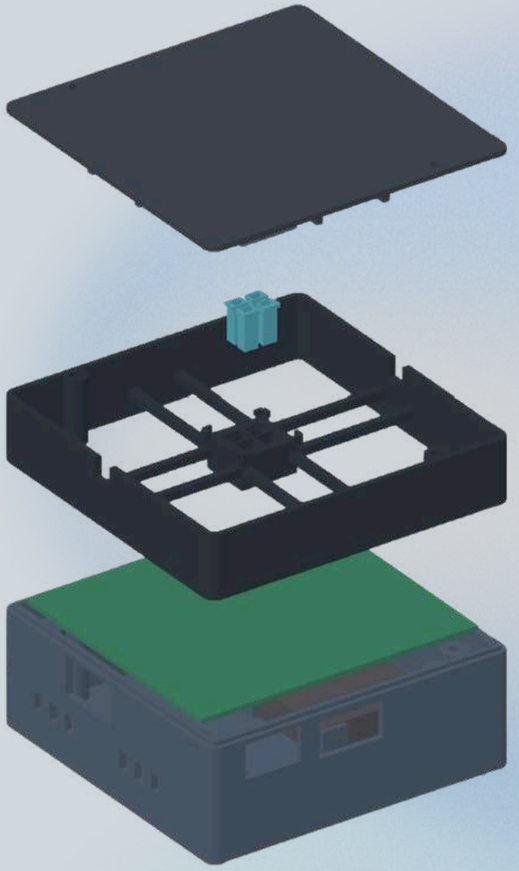
ESP32
lock-in recovers 4 clean channels

Like hearing a friend's code word at a loud party: everything off-frequency is filtered out.

Four channels at once, no moving filters, and room light and low-frequency drift are rejected.

HARDWARE · THE DEVICE

One box, from cells to verdict



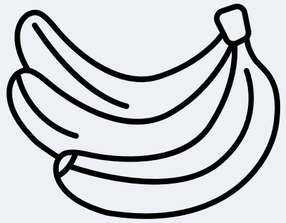
Lid	Gas in / gas out ports bring sample air over the cells
Cell cartridge	Four wells of receptor cells, one per channel
Signal board	Custom PCB: LEDs, photodiode, amplifier and ESP32 lock-in
Raspberry Pi	Power, control and the fly-brain decoder

Whole pipeline under 1 W, at the edge

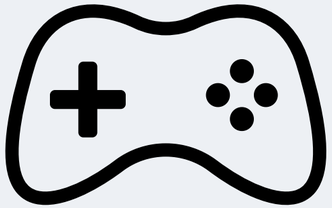
HUMAN PRACTICES

Who shaped it, and what changed?

Education



collab with banana expert



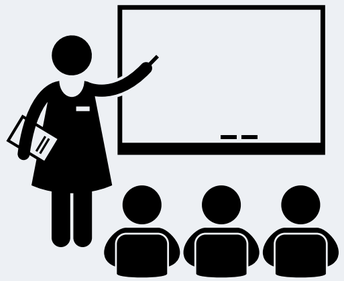
online games



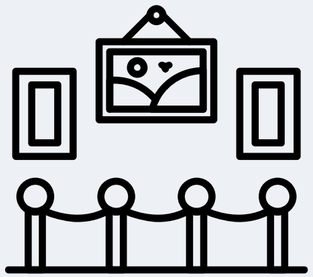
learning platform



neuroAI workshop



elementary camp



art gallery

Entrepreneurship

NTHU STARTUP GARAGE

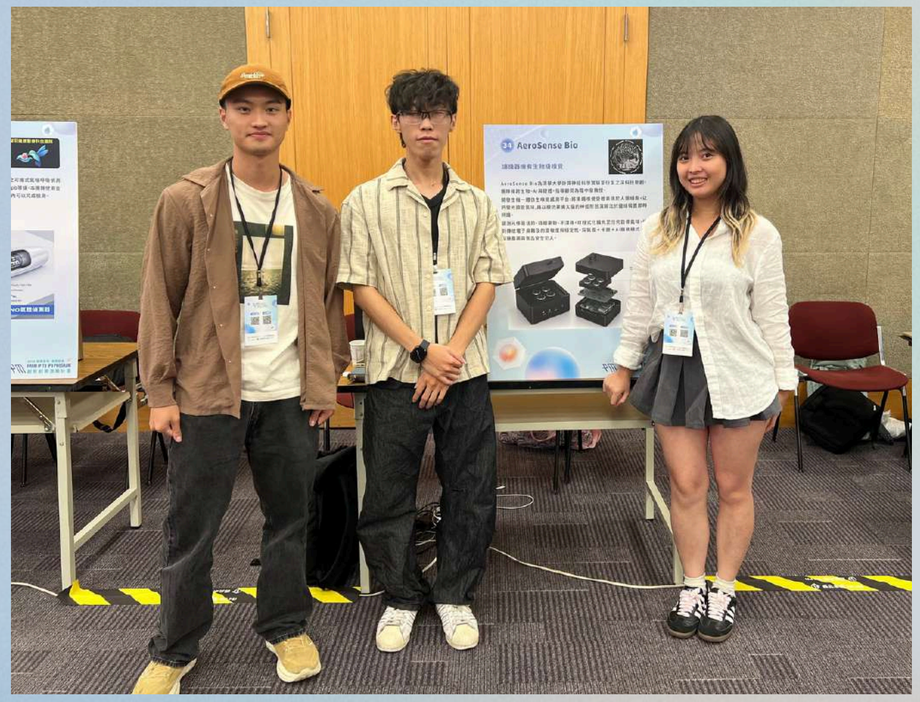




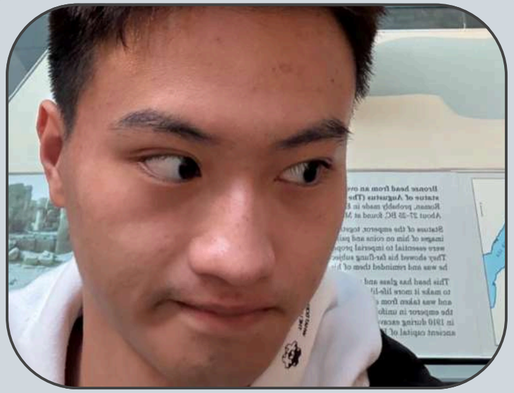
Entrepreneurship

FITI (From IP To IPO) Program by NSTC





Meet Our Team



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Thank You

for your time
and attention

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